

Mehul Srivastava

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github.com/Mehulsri07 | Portfolio

Final-year Computer Science (AI/ML) student working towards DevOps and SRE roles. Hands-on with Docker, GitHub Actions CI/CD, AWS EC2 and Prometheus/Grafana monitoring, and maintains a released desktop application used day to day by a real drilling business.

Education

Manipal University Jaipur 2023 – 2027
B.Tech, Computer Science & Engineering (AI/ML) *Jaipur, India*
- Relevant coursework: Operating Systems, Computer Networks, DBMS, Algorithms, AI/ML, NLP

Seth M.R. Jaipuria School 2008 – 2023
Class XII (ISC) – 80.25% (2023) | *Class X (ICSE) – 91.20% (2021)* *Lucknow, India*

Technical Skills

Languages: Python
Cloud & DevOps: AWS (EC2, Systems Manager), Docker, Docker Compose, GitHub Actions, GHCR, Kubernetes (kind), Linux (Ubuntu)
Observability & Testing: Prometheus, Grafana, Node Exporter, k6
Frameworks & Data: React, Tauri, PostgreSQL, SQLite, TensorFlow/Keras, Leaflet, Google Earth Engine

Projects

StrataField – Offline-First Desktop App for Borewell Data | *Tauri 2, React, SQLite* Released v1.0.2
- Built and shipped a Windows desktop app used by my family's drilling business to record borewells, soil layers, pipes, water levels and site photos, with an offline city map of water depths and geological cross-sections.
- Rebuilt the original Electron version on Tauri 2, cutting the installer from **134 MB to 3.5 MB**, and replaced sql.js with a native SQLite database (schema migrations, edit history, recycle bin, daily backups) that imports the old app's data automatically.
- Set up GitHub Actions CI with typecheck, lint, unit tests, an end-to-end test on Windows and a clean-machine installer test; tagged releases publish signed builds that installed copies auto-update from.

LoadLab – Load-Testing & Observability Sandbox | *Docker Compose, Prometheus, Grafana, k6* AWS EC2
- Deployed a Docker Compose stack on EC2: Nginx in front of a backend API with deliberately CPU- and memory-heavy routes, scraped by Prometheus alongside Node Exporter and visualised in Grafana.
- Wrote low/medium/high k6 load profiles to find where p95 latency climbs and which resource saturates first; fixed a Grafana redirect loop behind the Nginx sub-path proxy by correcting the root URL and forwarded headers.

IntelliOps – Agentic AIOps Platform | *Python, GitHub Actions, Docker* Team project (contributor)
- Contributed the CI/CD workflows (lint, tests, Docker Compose smoke test, frontend build gate, GHCR image builds) to a team-built platform for alert correlation, root-cause analysis and human-approved Kubernetes remediation.
- Added Kafka as an event bus interchangeable with the in-memory backend (checked by contract tests), a Helm chart for Postgres, Redis, all services and a migration job, and a load-test script.

Experience

India Space Academy May 2026 – Jul 2026
Remote Sensing & GIS Trainee *Remote*
- Produced remote sensing reports on the Urban Heat Island effect and Aerosol Optical Depth / NO2 over Lucknow using MODIS and Sentinel-5P data, processed in Google Earth Engine with band math and temporal compositing.

Publications

Srivastava M., Talwar M., Singh S. – "Deep Learning-Based Invisible Image Watermarking Using DWT and CNN" – IEEE (under review). Code: TensorFlow/Keras, evaluated with PSNR, SSIM and bit error rate under noise, JPEG, crop, scale and rotation attacks.

Leadership

Aperture Digital Arts Society, Manipal University Jaipur
Head of Coverage Photography
- Led a photography team across 75+ event assignments, including CEO Conclave, Manipal Marathon, Oneiros and TechIdeate, managing coverage, delivery timelines and membership drives.